

Skeletal muscle relaxants

By: Dr. Riffat Ibrahim Riffat



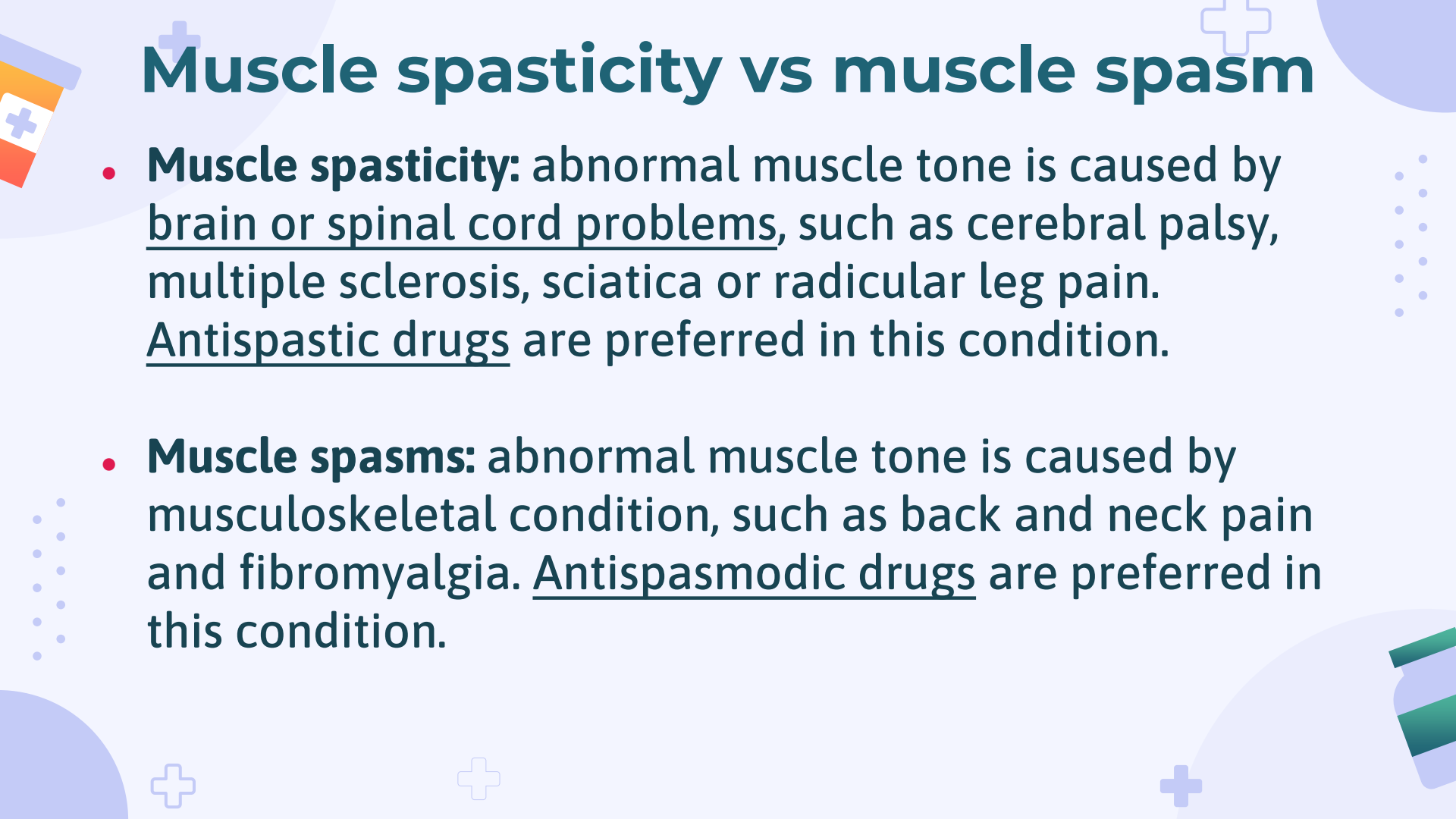
A large, light purple stethoscope graphic is centered in the background, with its chest piece and earpieces visible.

01 overview



Introduction

- Muscle relaxants are two major groups, neuromuscular blockers (NMBs) and spasmolytics.
 - Neuromuscular blockers act by blocking nicotinic acetylcholine receptors at the neuromuscular junction and have no CNS activity, they are often used during surgical procedures and in intensive care.
 - Spasmolytics: are centrally acting muscle relaxants, are used to relieve spasms (musculoskeletal pain) and spasticity (neurological conditions)



Muscle spasticity vs muscle spasm

- **Muscle spasticity:** abnormal muscle tone is caused by brain or spinal cord problems, such as cerebral palsy, multiple sclerosis, sciatica or radicular leg pain. Antispastic drugs are preferred in this condition.
- **Muscle spasms:** abnormal muscle tone is caused by musculoskeletal condition, such as back and neck pain and fibromyalgia. Antispasmodic drugs are preferred in this condition.

The background is a light blue gradient. A large, faint stethoscope is centered behind the text. In the top left, there is a clipboard with a checklist and a white plus sign. In the top right, there is a red and white pill bottle. In the bottom left, there is a green pill bottle with a white plus sign. In the bottom right, there is a red heart with a white plus sign. Several small white plus signs are scattered throughout the background.

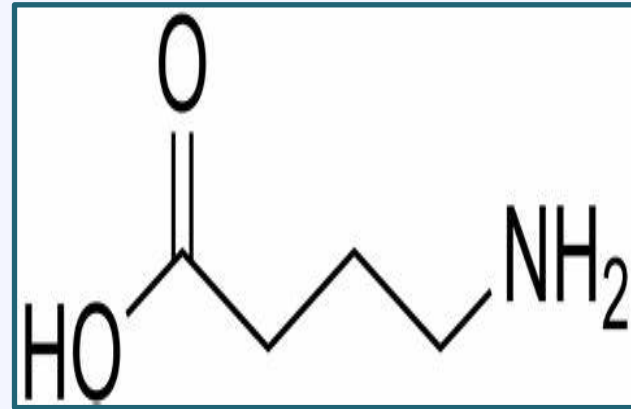
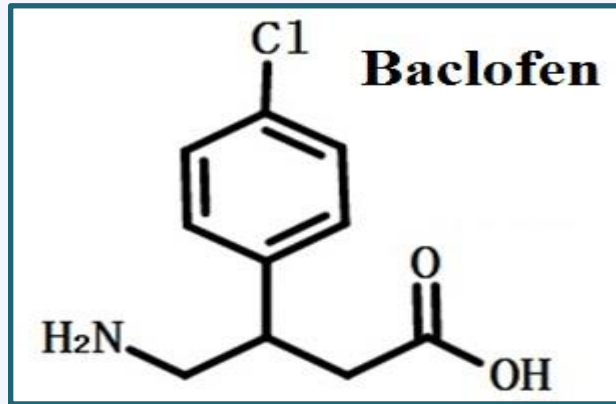
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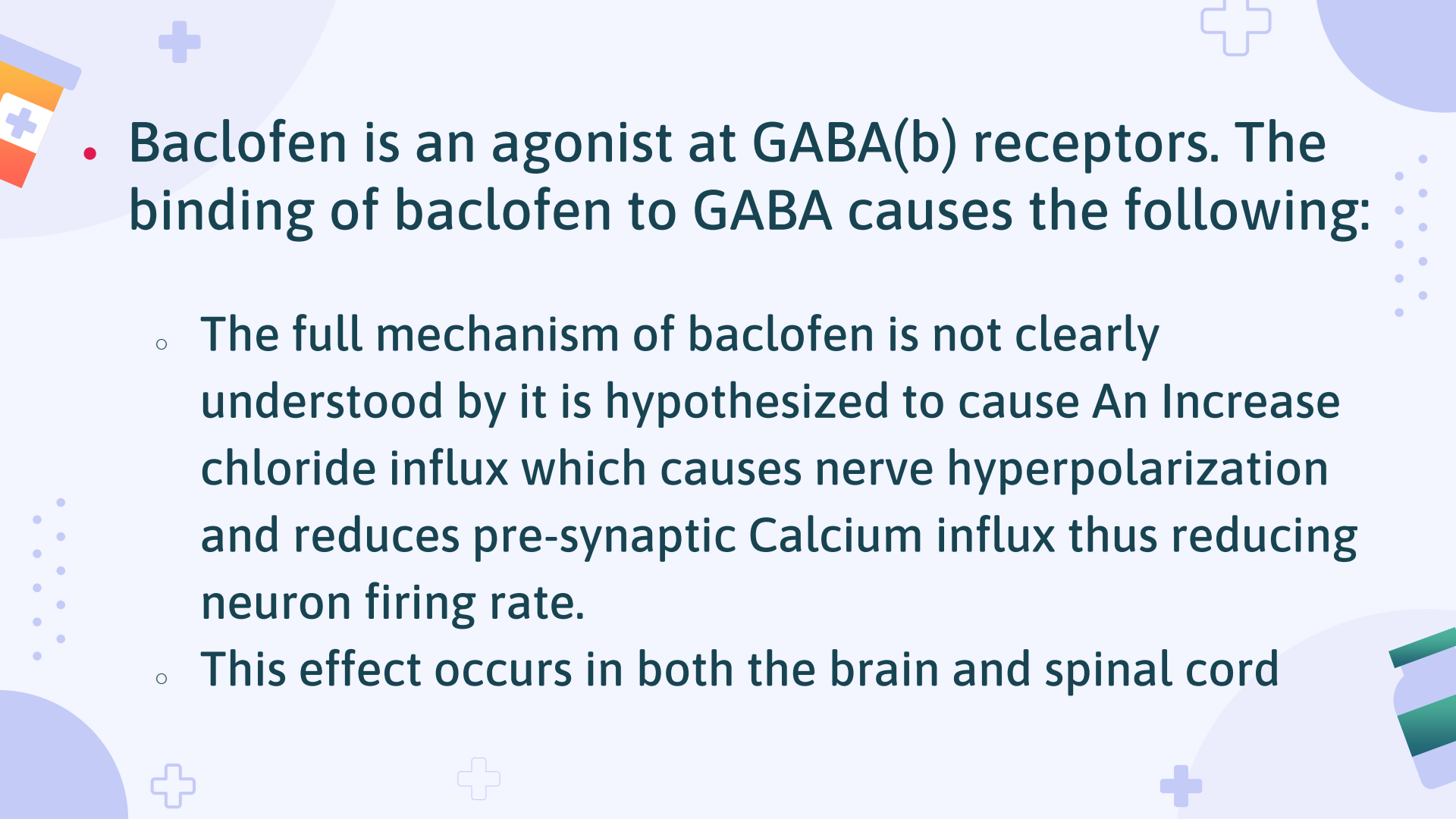
Central antispasticity agents

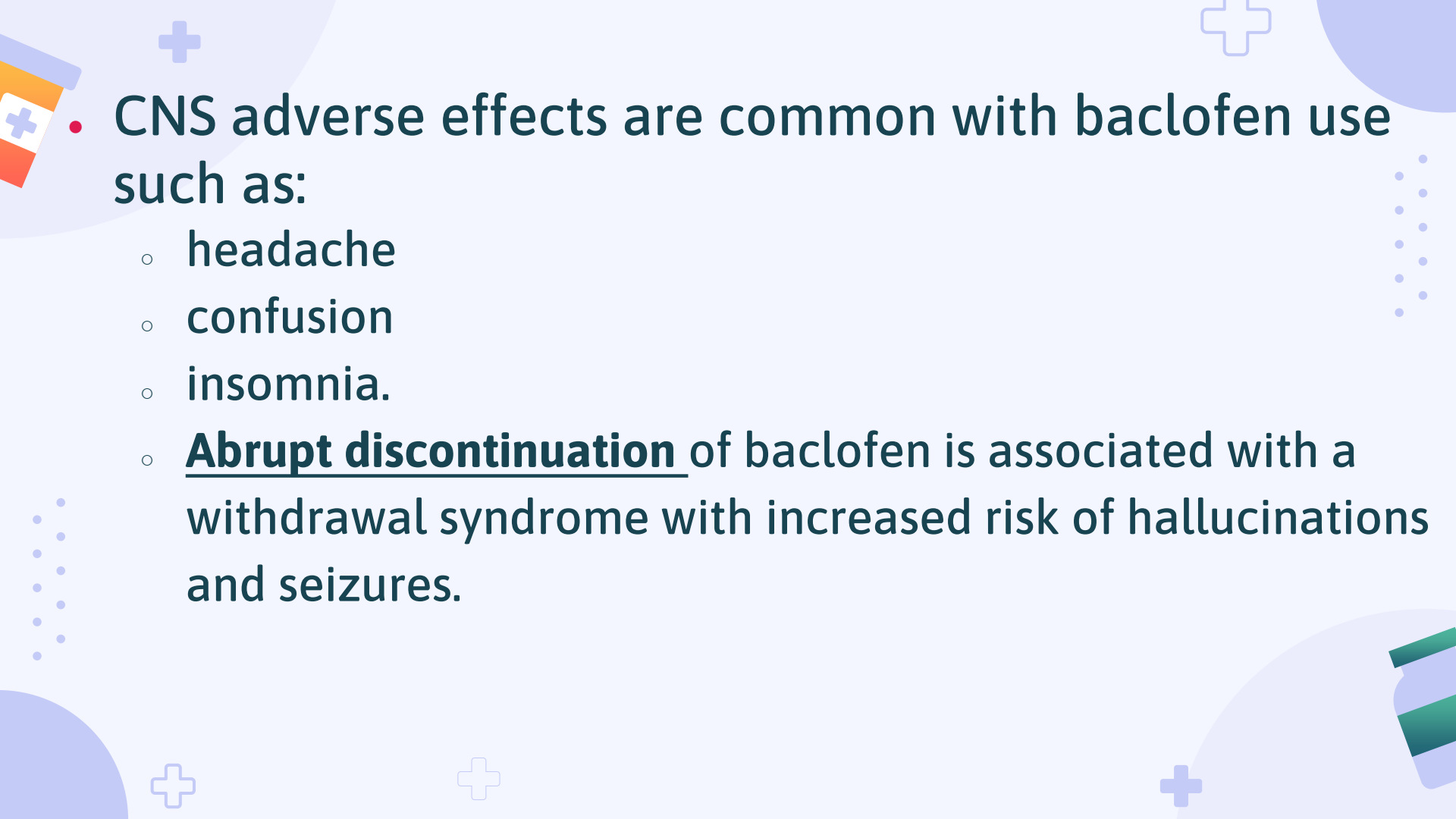
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- **Spasticity** is a disruption in muscle movement patterns that causes certain muscles to contract all at once when you try to move or even at rest. It's usually caused by damage to nerve pathways within your brain or spinal cord that control movement and stretch reflexes.
 - In general, antispastic medications act on your spinal cord or skeletal muscle directly to improve muscle tightness (hypertonicity) and involuntary spasms.
 - Certain conditions like: multiple sclerosis, spinal cord injury, cerebral palsy greatly benefit from antispastic drugs to alleviate muscle contraction and pain.
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Baclofen

- Baclofen is a derivative of GABA (chlorophenyl-GABA) that was designed to be an orally active GABA- potentiating agent.



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- Baclofen is an agonist at GABA(b) receptors. The binding of baclofen to GABA causes the following:
 - The full mechanism of baclofen is not clearly understood by it is hypothesized to cause An Increase chloride influx which causes nerve hyperpolarization and reduces pre-synaptic Calcium influx thus reducing neuron firing rate.
 - This effect occurs in both the brain and spinal cord

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- CNS adverse effects are common with baclofen use such as:
 - headache
 - confusion
 - insomnia.
 - **Abrupt discontinuation** of baclofen is associated with a withdrawal syndrome with increased risk of hallucinations and seizures.

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Tizanidine

Tizanidine is centrally acting α_2 agonist that is used as an antispastic agent.

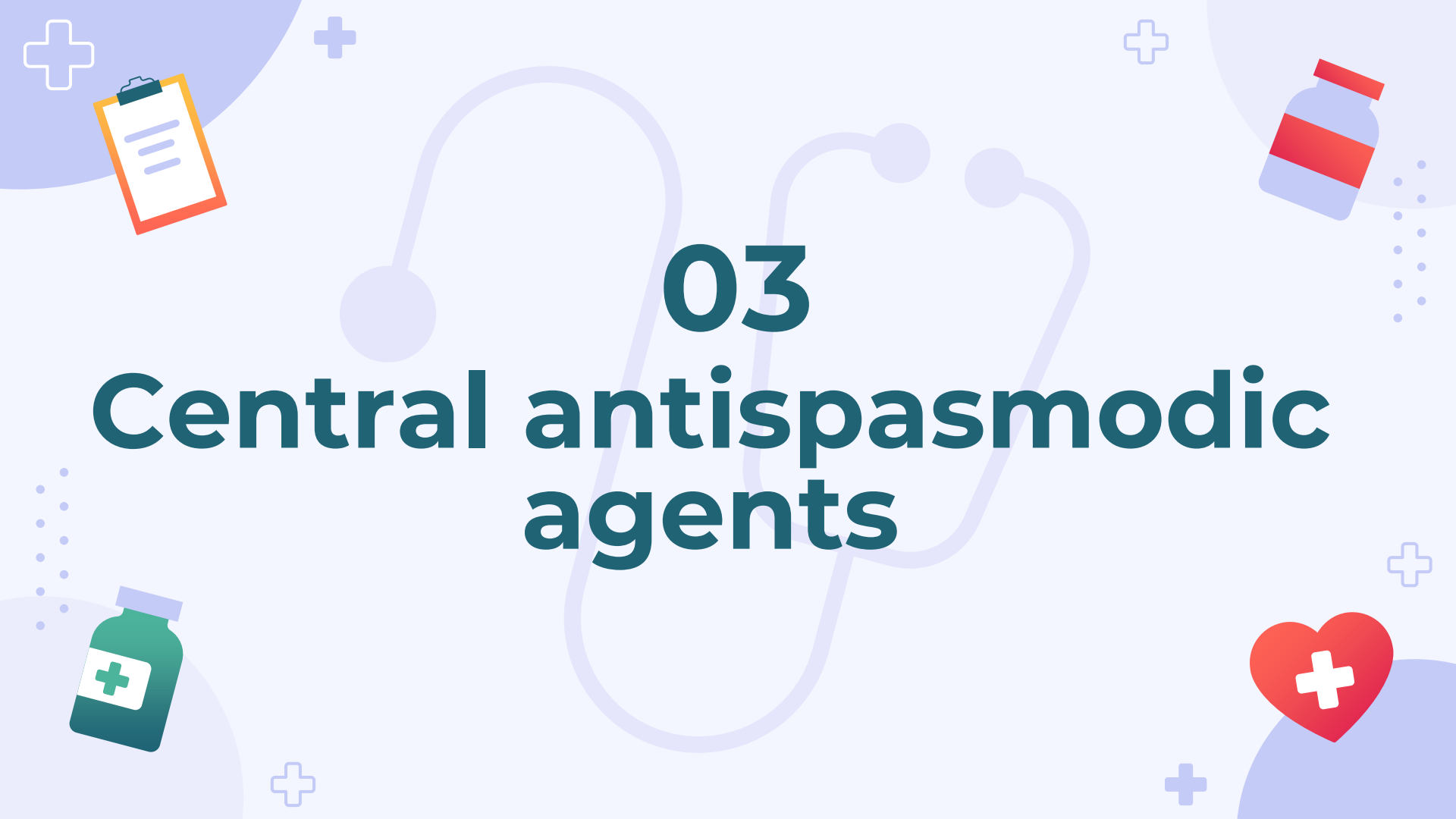
- Tizanidine is used to treat spasticity and spasm at doses much lower than the dose that would produce a cardiovascular effect.

Adverse effects are dose related such as:

- Hypotension, dry mouth, and sedation (common with α_2 agonist) .
- Abrupt withdrawal will cause rebound hypertension.
- Tizanidine is metabolized by CYP-1A2 and inhibitors of this enzyme (ciprofloxacin) will cause tizanidine toxicity.

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Central antispasmodic agents

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- Providers mainly prescribe **antispasmodic skeletal muscle relaxants for musculoskeletal and myofascial pain, especially lower back pain, and muscle spasms.**
 - these medications aren't the first-line therapy (NSAIDs, physiotherapy, and lifestyle adjustment are used first) for the treatment of pain or spasms.
 - This is due to the adverse side effects of antispasmodics, including addiction potential for some types of the medication.

Cyclobenzaprine

- it is structurally related to TCA with a similar mechanism of action and adverse effects.
- It used for short term management of muscle spasm associated with acute musculoskeletal conditions and off-label for fibromyalgia.
- Important notes:
 - It should be used with caution in the elderly and hepatic dysfunction.
 - Discontinuation of cyclobenzaprine for 14 days is required before initiating MAOI therapy
 - Serotonin syndrome is a risk when combined with a serotonergic drug such as (SSRI, sumatriptan, ergot alkaloids)

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Orphenadrine

Orphenadrine is an anticholinergic drug closely related to the 1st generation antihistamine [Diphenhydramine].

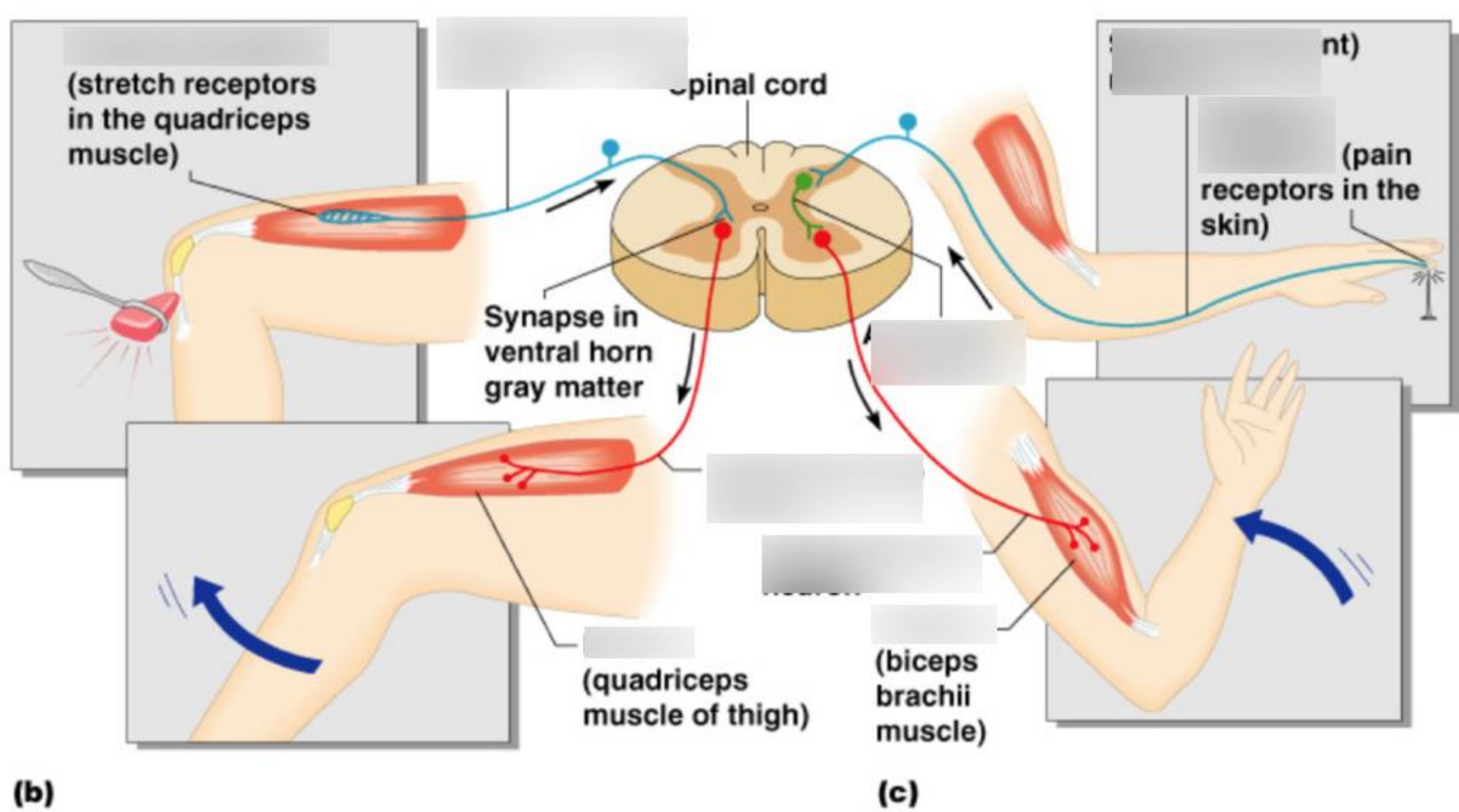
- It is used in combination with paracetamol for treatment of muscle spasm conditions. its mechanism of action in causing analgesia and skeletal muscle relaxation is not well defined.
- Its main adverse effects are anticholinergic side effects.
- orphenadrine can be abused in Large doses to cause hallucinations.

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Chlorzoxazone

- The mechanism of action is not fully understood, but it works primarily in spinal cord by inhibiting multi-synaptic reflex arcs involved in producing and maintaining skeletal muscle spasm.
- Chlorzoxazone is available in combination with analgesics such as ibuprofen or paracetamol.
- Chlorzoxazone should be used only for short term management espically when combined with paracetamol due to a risk of fatal hepatocellular toxicity.



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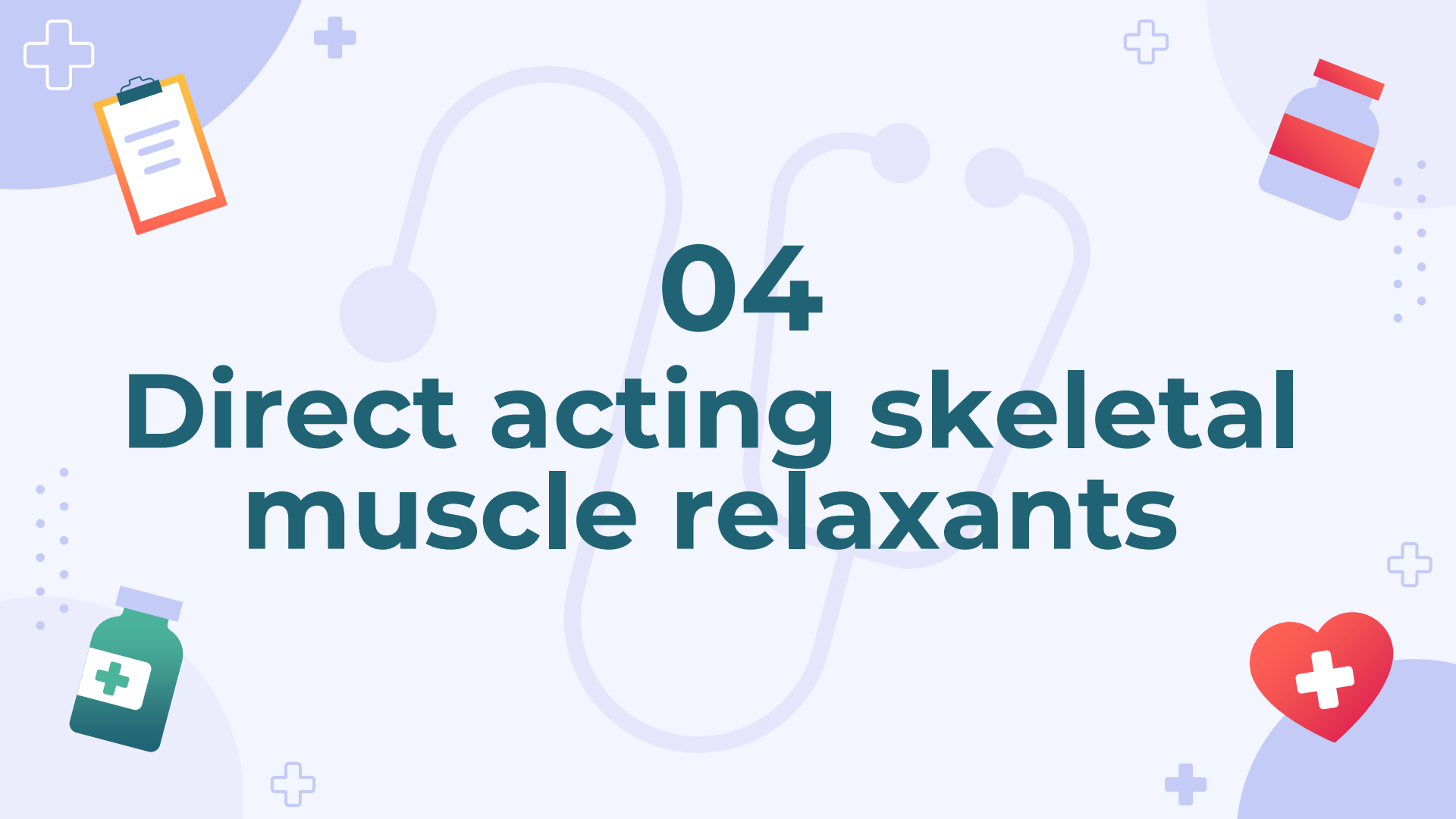


Methocarbamol

- Methocarbamol is carbamate derivative of guaifenesin. And is used in a manner similar to chlorzoxane.
- The mechanism of action of methocarbamol is not fully understood, however, it is used due to having the lowest abuse potential of all antispasmodics.
- Possible side effects are: dizziness, lightheadedness, and drowsiness. Methocarbamol is a well tolerated drug.

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Direct acting skeletal muscle relaxants

Dantrolene

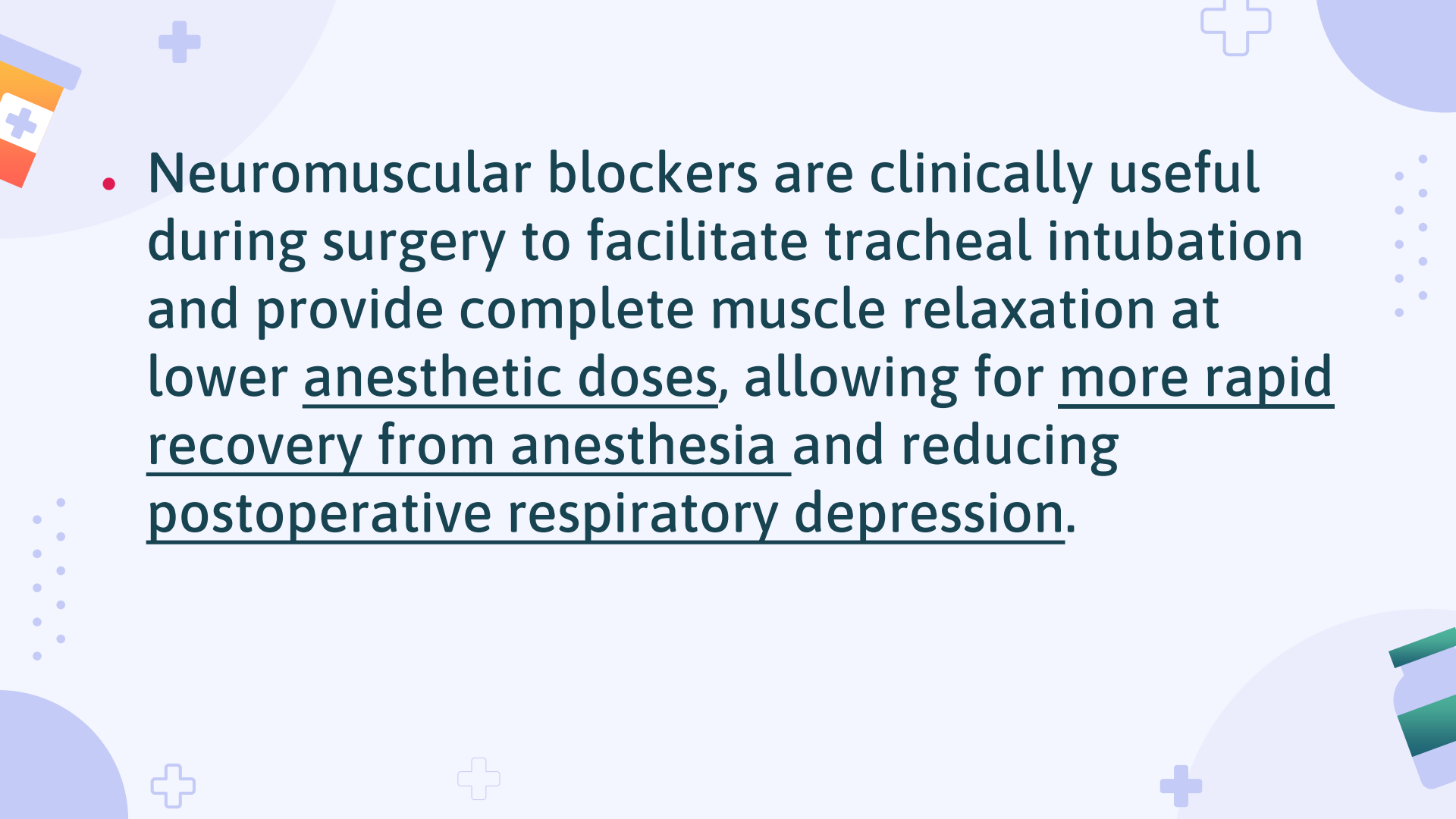
- Dantrolene is a hydration derivative related to phenytoin that has a unique peripheral mechanism of spasmolytic activity.
- The mechanism of action for dantrolene has a unique mechanism in which it inhibits Calcium ions release from sarcoplasmic reticulum.
- Dantrolene is used for treatment of various chronic muscle spasticity conditions such as MS, and spinal cord injury.
- Dantrolene (I.V) is also used for management of muscle rigidity associated with malignant hyperthermia (40 C) [a severe reaction to certain drugs used for anesthesia caused by genes].
- Drowsiness and dizziness are common with dantrolene use, the drugs comes with a black box warning for hepatotoxicity.

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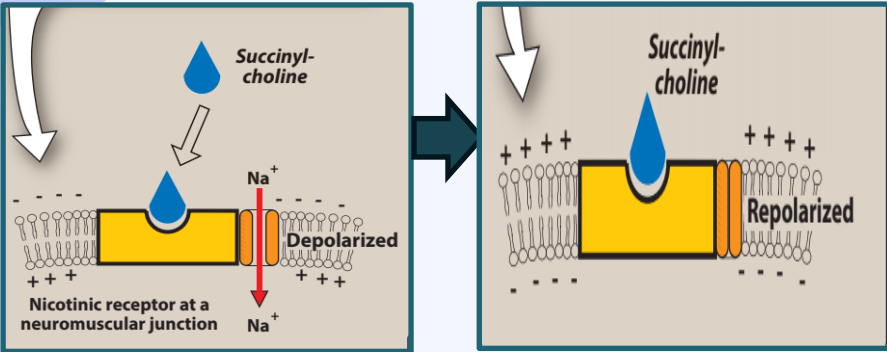


05 Neuromuscular blockers



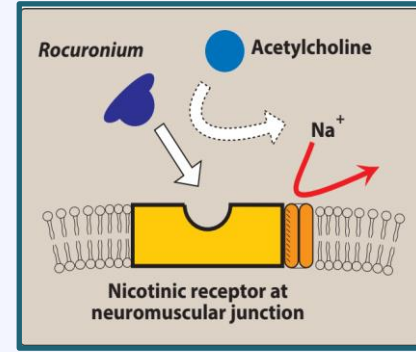
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- Neuromuscular blockers are clinically useful during surgery to facilitate tracheal intubation and provide complete muscle relaxation at lower anesthetic doses, allowing for more rapid recovery from anesthesia and reducing postoperative respiratory depression.

Classification



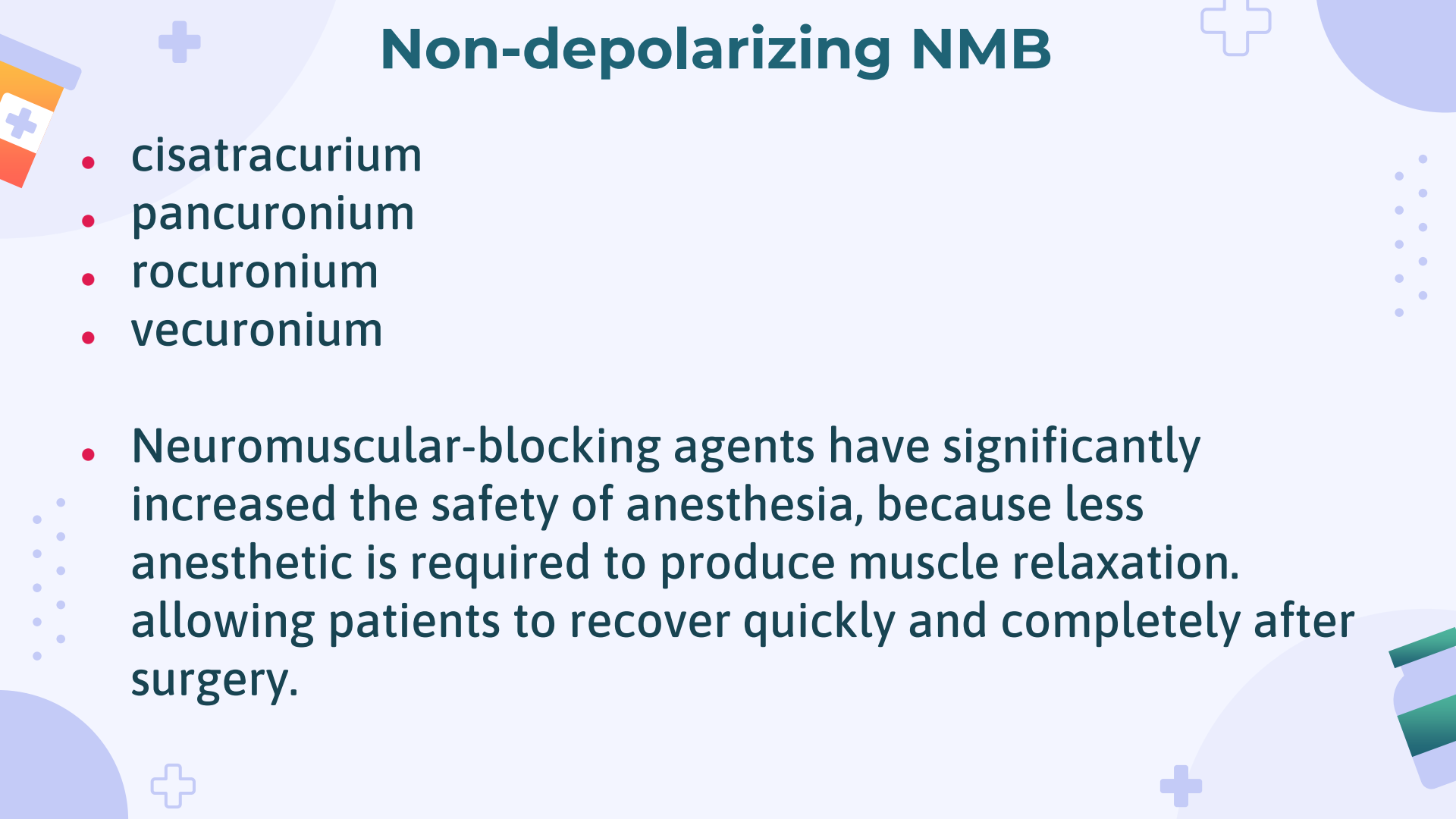
Depolarizing NMB

These drugs block cholinergic transmission between motor nerve endings and the nicotinic receptors on the skeletal muscle by constant activation of the receptor which causes a state of hyperpolarization. This state prevents muscle response to further stimulation.



Non-depolarizing NMB

These drugs block cholinergic transmission between motor nerve endings and the nicotinic receptors on the skeletal muscle by simply occupying the Nm receptor without affecting nerve polarization.

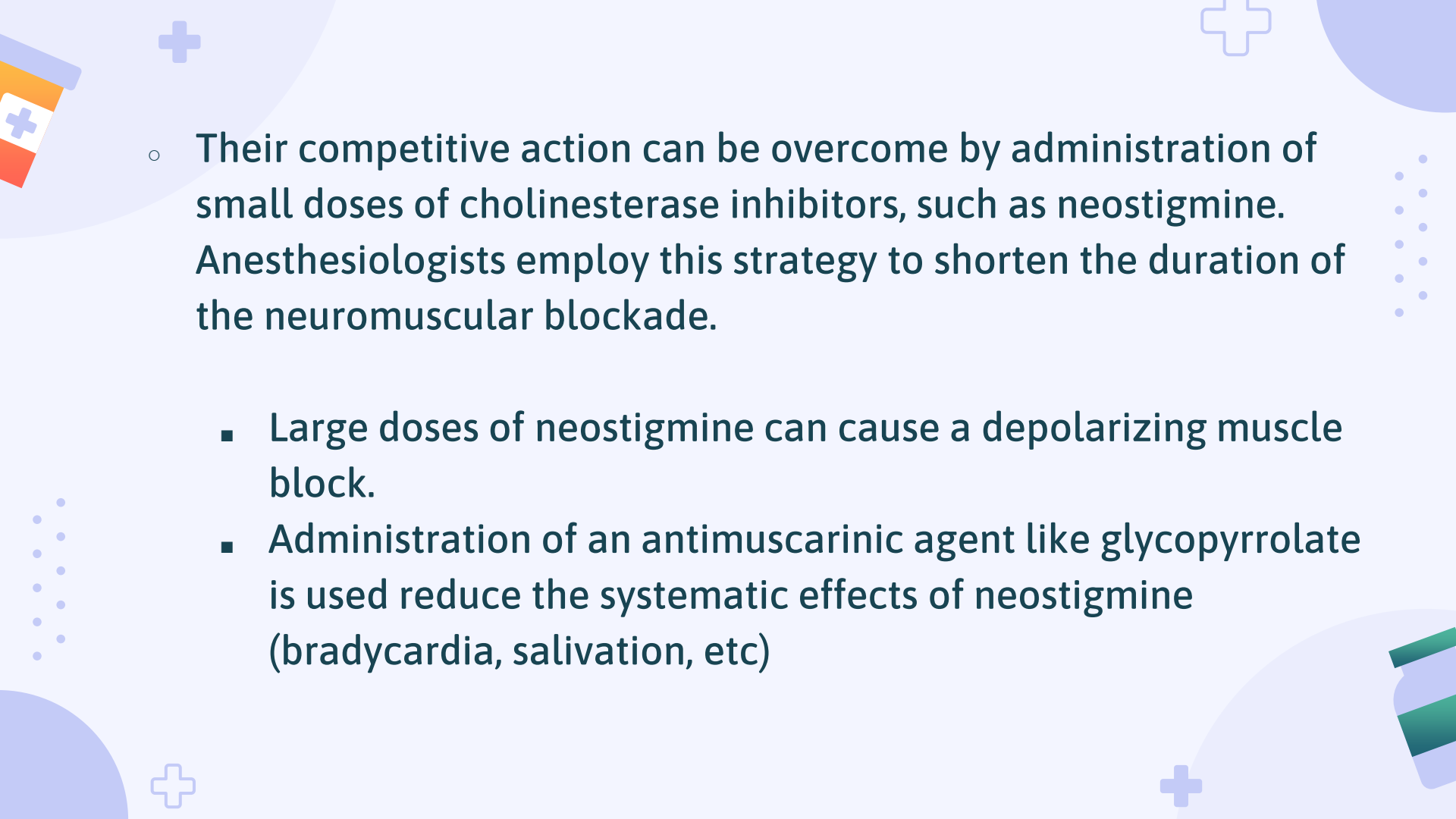


Non-depolarizing NMB

- cisatracurium
 - pancuronium
 - rocuronium
 - vecuronium
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- Neuromuscular-blocking agents have significantly increased the safety of anesthesia, because less anesthetic is required to produce muscle relaxation. allowing patients to recover quickly and completely after surgery.

- Mechanism of action:

- Nondepolarizing agents competitively block ACh at the nicotinic receptors without stimulating it. Thus, these drugs prevent depolarization of the muscle cell membrane and inhibit muscular contraction.
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- Their competitive action can be overcome by administration of small doses of cholinesterase inhibitors, such as neostigmine. Anesthesiologists employ this strategy to shorten the duration of the neuromuscular blockade.
 - Large doses of neostigmine can cause a depolarizing muscle block.
 - Administration of an antimuscarinic agent like glycopyrrolate is used reduce the systematic effects of neostigmine (bradycardia, salivation, etc)

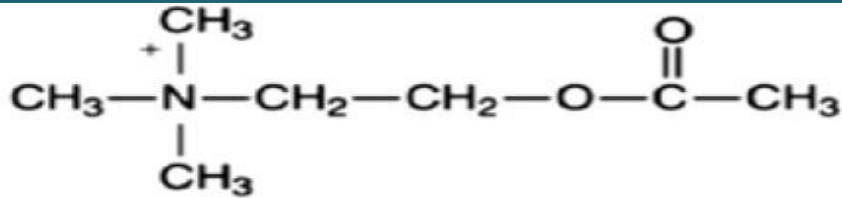


Pharmacokinetics:

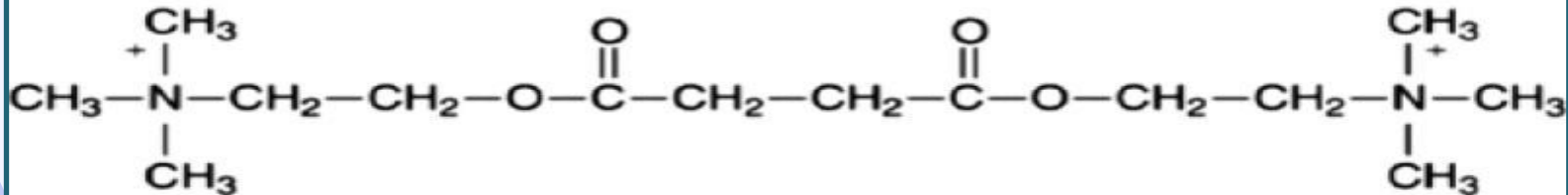
- These agents are administered I.V as they are very polar and can not be absorbed orally. They also can not cross B.B.B
 - The choice of an agent depends on the desired onset and duration of the muscle relaxation and route of elimination.
 - vecuronium is hepatically metabolized to inactive metabolites. (not suitable for hepatic dysfunction)
 - pancuronium is eliminated renally unchanged (not suitable for kidney failure)
 - cisatracurium is metabolized in plasma and can be used in patients with multiple organ failure.
 -  In general, these agents are safe with minimal side effects.
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Depolarizing NMB

- Succinylcholine is the only member of this family used in clinical application, it is structurally similar to acetylcholine, however, it is resistant to synaptic acetylcholinesterase
- Succinylcholine is administered I.V where almost 90% of the administered dose is metabolized through hydrolysis by plasma pseudocholinesterase.



Acetylcholine



Succinylcholine

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- mechanism of action:
 - It's mode of action occurs in two phases:
 - Phase 1: activation of Nm receptors causes the opening of the sodium channel associated with the nicotinic receptors, which results in depolarization of the receptor. This leads to a transient twitching of the muscle.
 - Phase 2: Continued binding of the depolarizing agent renders the receptor incapable of transmitting further impulses. With time, continuous depolarization gives way to gradual repolarization as the sodium channel closes or is blocked. This causes a resistance to depolarization and flaccid paralysis.
 - N.B [phase 2 occurs as result of synaptic acetylcholine esterase not being able to breakdown succinylcholine.]
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Adverse effects :

- **Malignant hyperthermia:** in a very small percentage of susceptible patients, exposure to halogenated hydrocarbon anesthetics or the neuromuscular blocker succinylcholine may induce malignant hyperthermia (MH)
- [a rare life-threatening condition. MH causes a drastic and uncontrolled increase in skeletal muscle oxidative metabolism, overwhelming the body's capacity to supply oxygen, remove carbon dioxide, and regulate temperature, eventually leading to circulatory collapse and death if not treated immediately.]
- H. Susceptibility to MH is often inherited

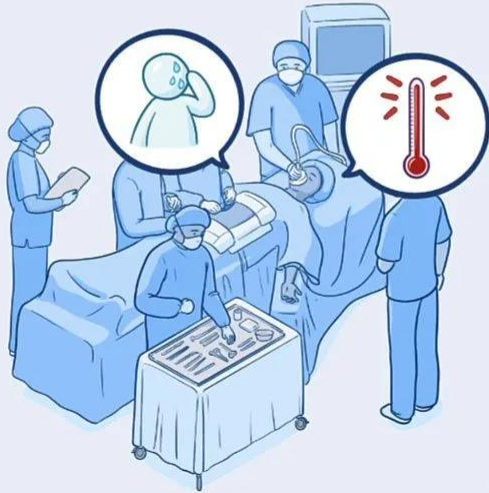


MALIGNANT HYPERTHERMIA

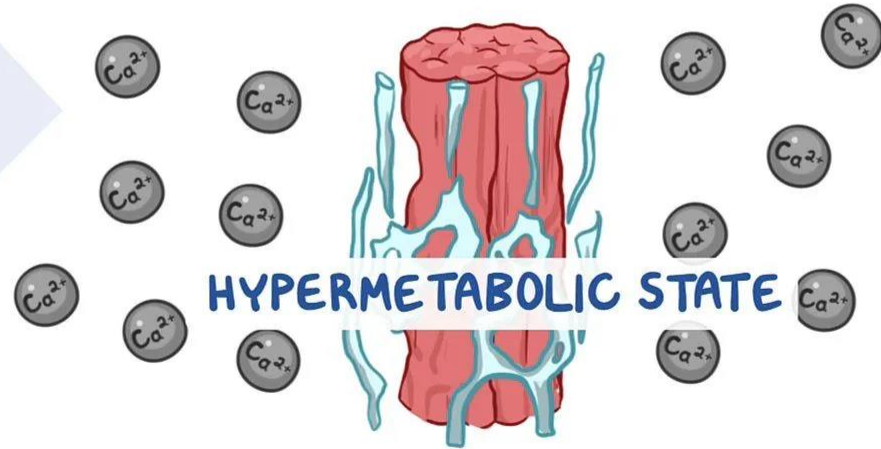
SERIOUS ANESTHETIC COMPLICATION

GENETIC MUTATION

* DIHYDROPYRIDINE
or RYANODINE
RECEPTORS



CALCIUM OVERLOAD causes
SUSTAINED MUSCLE CONTRACTION
& BREAKDOWN

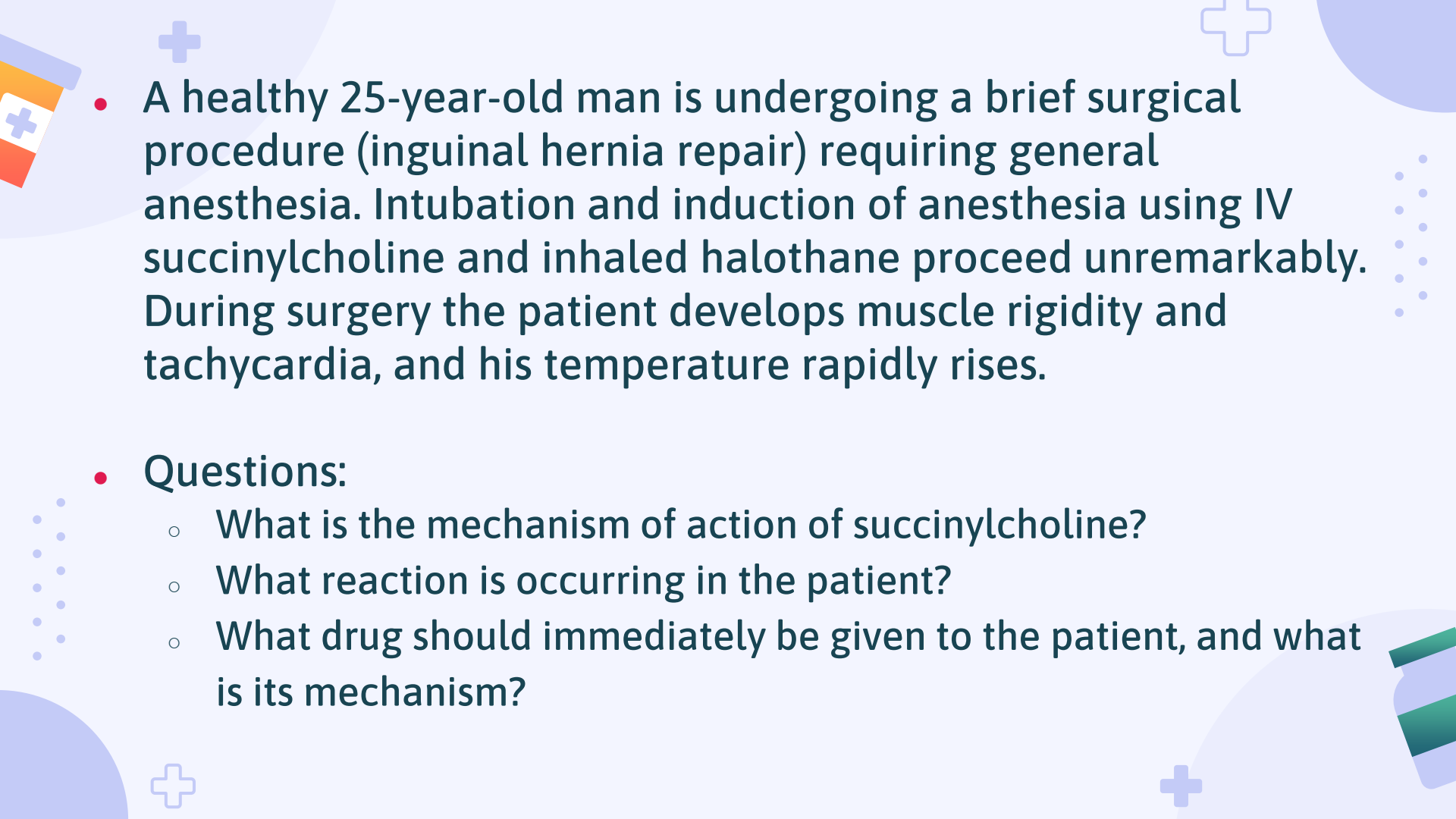


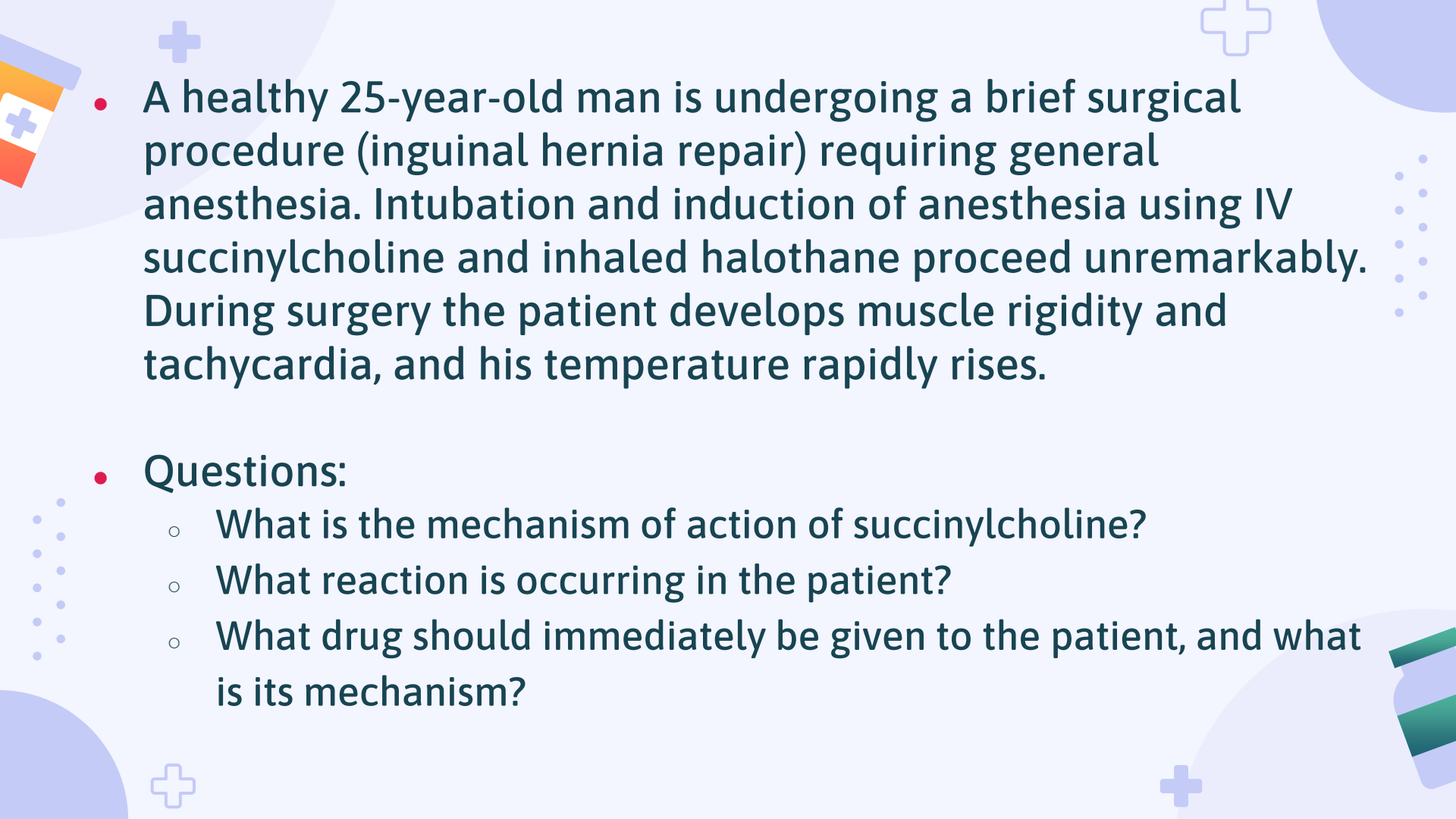
- **Apnea:** Administration of succinylcholine to a patient who is deficient in plasma cholinesterase or who has an atypical form of the enzyme can lead to prolonged apnea due to paralysis of the diaphragm.

A large, light purple stethoscope is centered in the background, with its chest piece and earbuds visible.

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Cases



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- A healthy 25-year-old man is undergoing a brief surgical procedure (inguinal hernia repair) requiring general anesthesia. Intubation and induction of anesthesia using IV succinylcholine and inhaled halothane proceed unremarkably. During surgery the patient develops muscle rigidity and tachycardia, and his temperature rapidly rises.

- Questions:
 - What is the mechanism of action of succinylcholine?
 - What reaction is occurring in the patient?
 - What drug should immediately be given to the patient, and what is its mechanism?
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• Answers:

- Nicotinic receptor agonist at the motor endplate of the neuromuscular junction, which causes persistent stimulation and depolarization of muscle cells.
 - Malignant hyperthermia.
 - Dantrolene, which acts by interfering with calcium release from the sarcoplasmic reticulum.
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